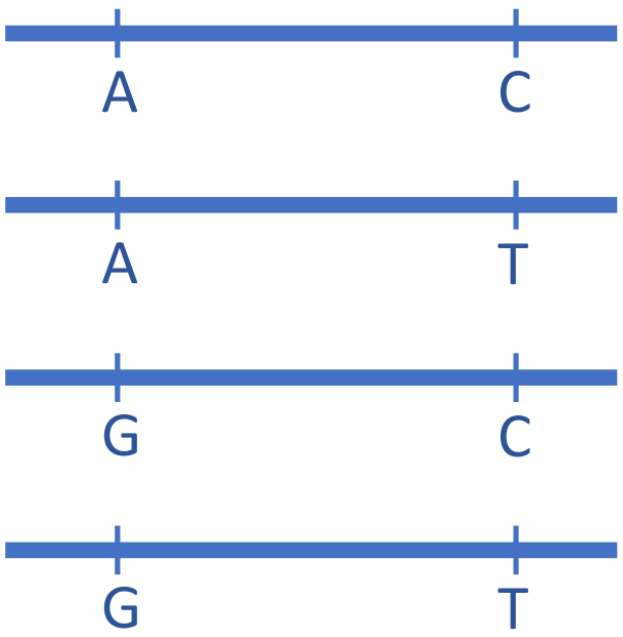
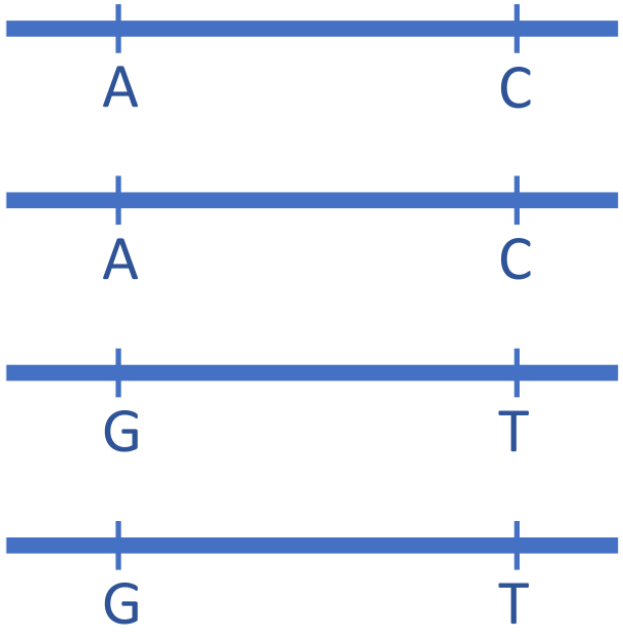
LD and LD pruning

Linkage disequilibrium (LD)

Nonrandom association of alleles at different loci



Equilibrium



Disequilibrium

What affects LD?

- Genome-wide
  - Population history
  - Breeding system
  - Pattern of geographic subdivision
- Within each genomic region
  - The history of natural selection, gene conversion, mutation and other forces that cause gene-frequency evolution
- Effect depends on local recombination rates (in complex interaction with selection, mutation and genetic drift)

Why estimate LD?

Why estimate LD?

- Identify genomic regions with elevated LD
- Examine the degree of LD between particular variants/regions of interest
- Understand LD decay patterns (and whether they are different between populations)
- Reconstruct recent effective population size histories
- Identify genomic blocks that may bias inference of population structure

Good resource for learning more

Nature Reviews Genetics, 9(6), 477–485 (2008)

FUNDAMENTAL CONCEPTS IN GENETICS

Linkage disequilibrium — understanding the evolutionary past and mapping the medical future

Montgomery Slatkin

Abstract | Linkage disequilibrium — the nonrandom association of alleles at different loci — is a sensitive indicator of the population genetic forces that structure a genome. Because of the explosive growth of methods for assessing genetic variation at a fine scale, evolutionary biologists and human geneticists are increasingly exploiting linkage disequilibrium in order to understand past evolutionary and demographic events, to map genes that are associated with quantitative characters and inherited diseases, and to understand the joint evolution of linked sets of genes. This article introduces linkage disequilibrium, reviews the population genetic processes that affect it and describes some of its uses. At present, linkage disequilibrium is used much more extensively in the

Different measures of LD

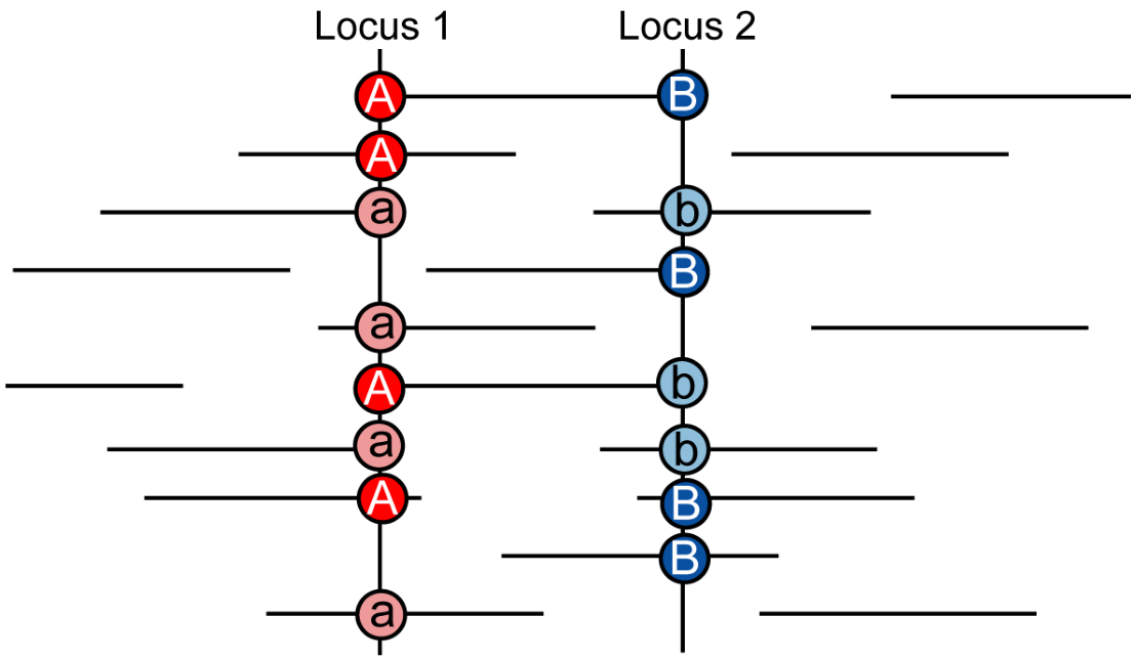
- Several different measures of LD but r^2 and D’ are most common

$$r^2 = \frac{D^2}{p_A(1-p_A)\,p_B(1-p_B)}$$ *(tests how well one site predicts the other)*

$$D = p_{AB} - p_A p_B \text{ with } D' = \frac{D}{D_{\max}}$$ (*measures occurrence of haplotypes)*

$$D_{\max} = \min(p_A(1-p_B), (1-p_A)p_B) \quad \text{when } D > 0$$
$$D_{\max} = \min(p_A p_B, (1-p_A)(1-p_B)) \quad \text{when } D < 0$$

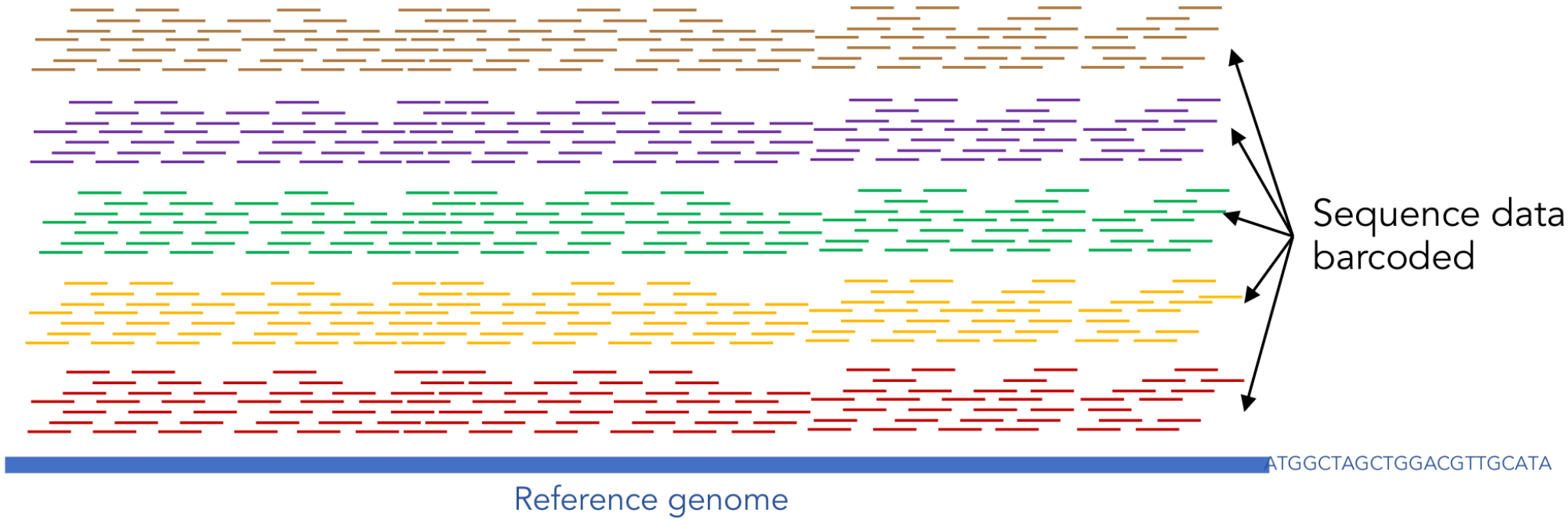
With Poolseq data, we can only estimate LD across the length of reads



Feder et al. 2012. PLOS ONE

Can only estimate LD between SNPs spanned by single read pairs (i.e. <1000bp)

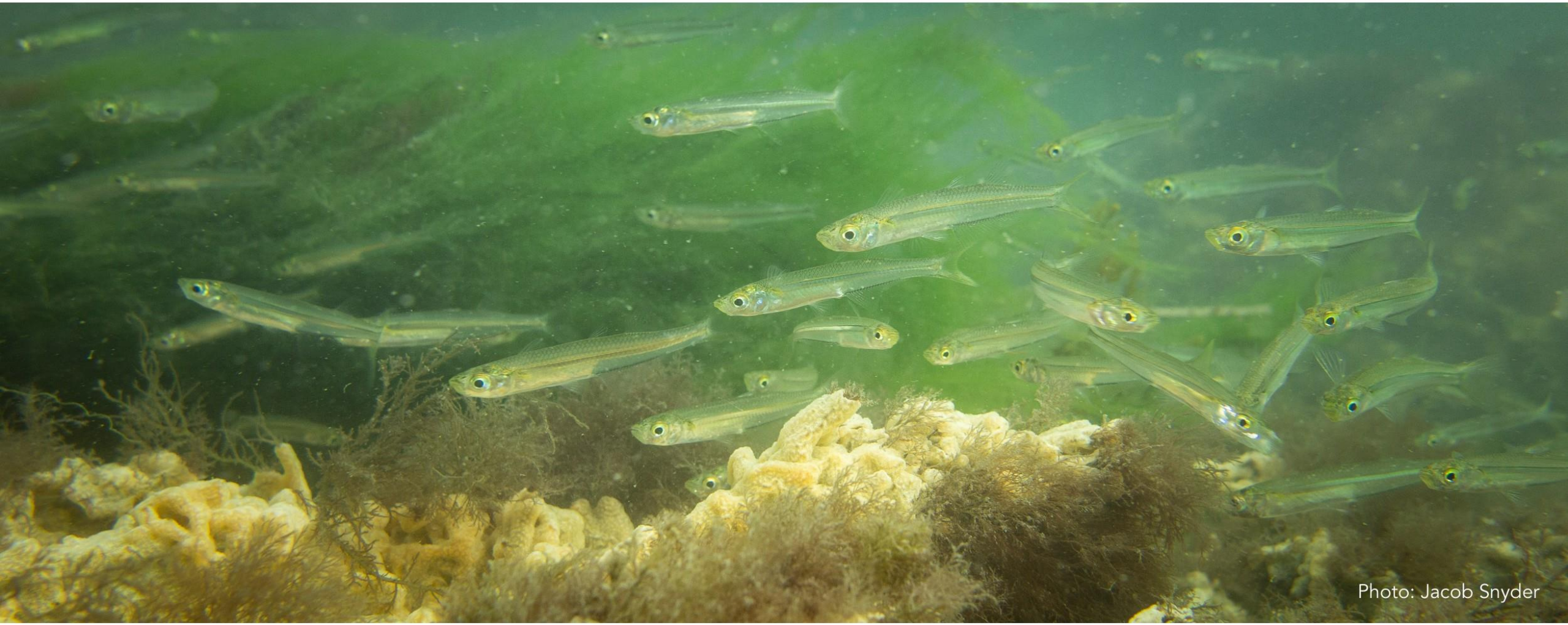
With lcWGS data, we preserve information about which individual carried each observed allele across the entire genome

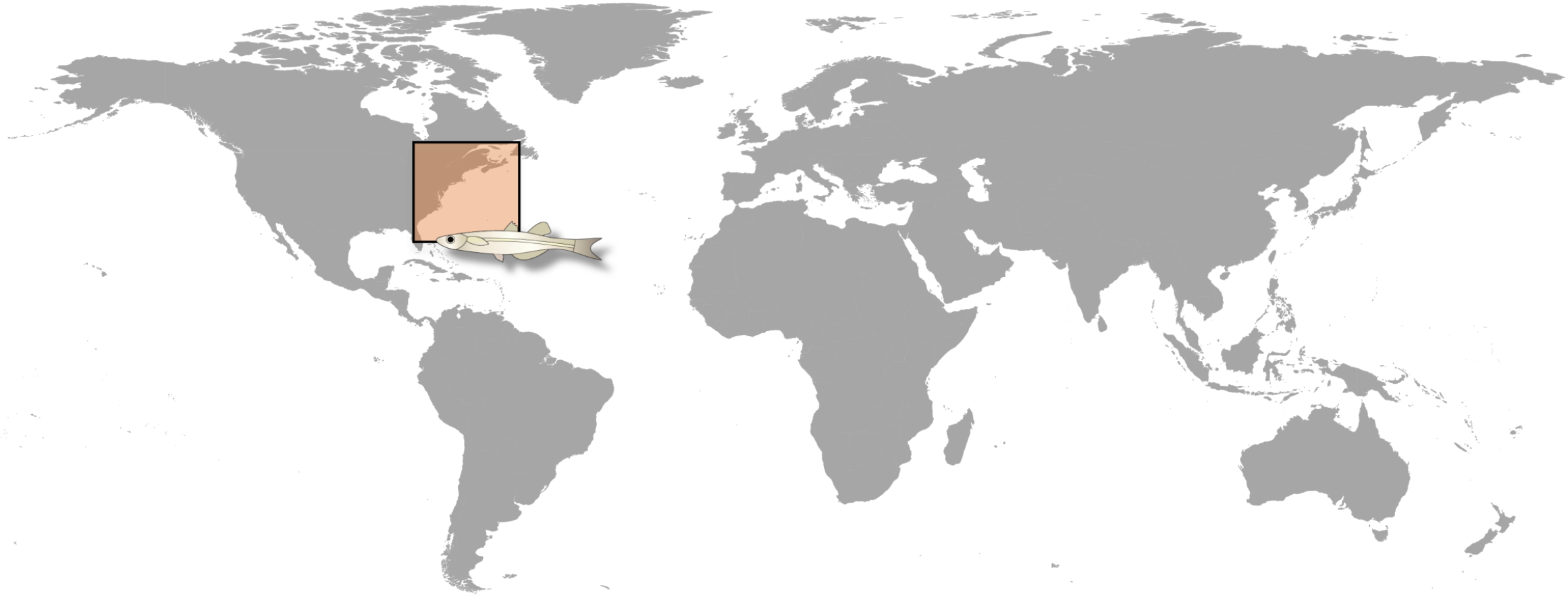


Adapted from https://sites.google.com/site/wiki4metagenomics/pdf/definition/coerage-read-depth

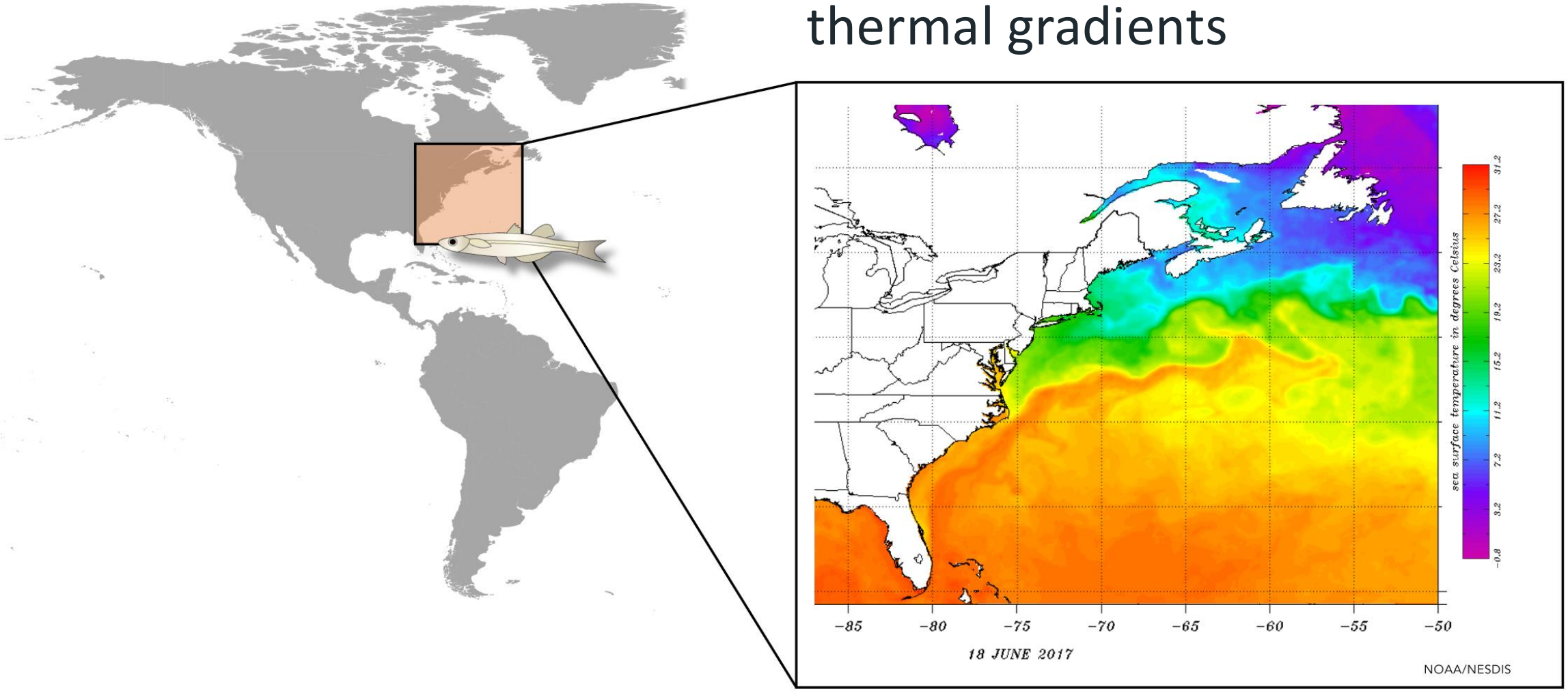
Some examples

Atlantic silverside *Menidia menidia*



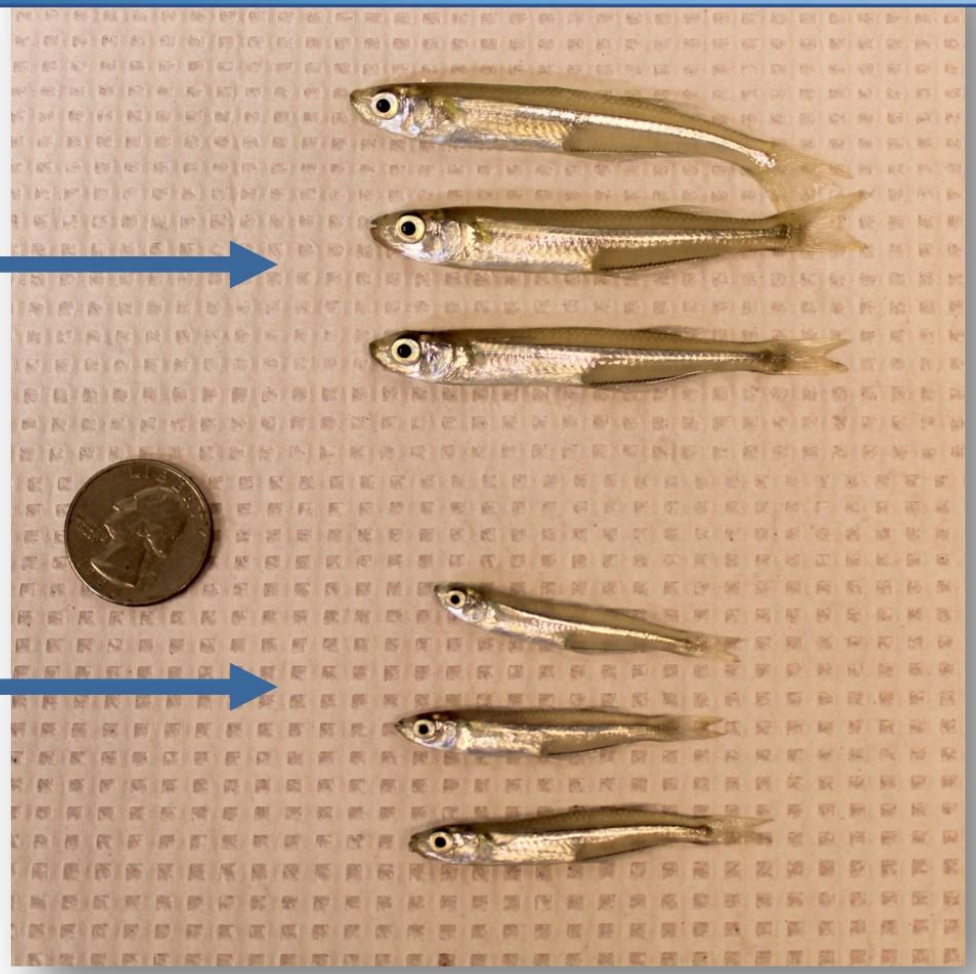


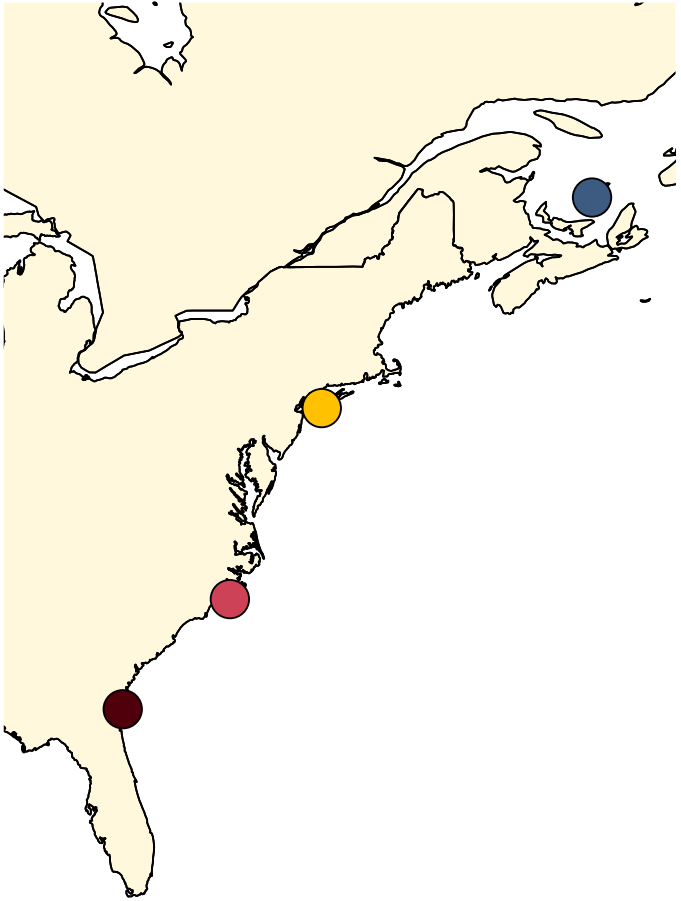
One of the world's steepest thermal gradients





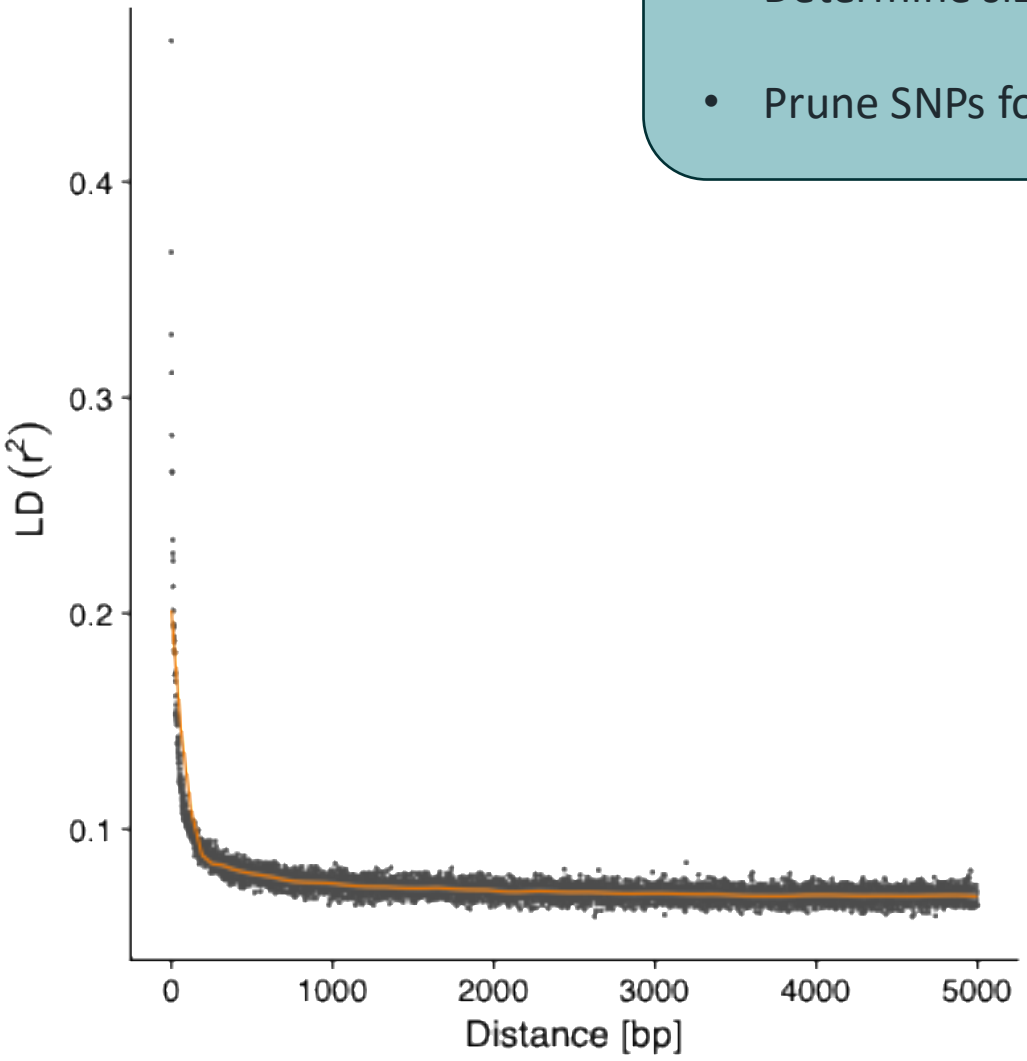
Same age,
Common lab environment



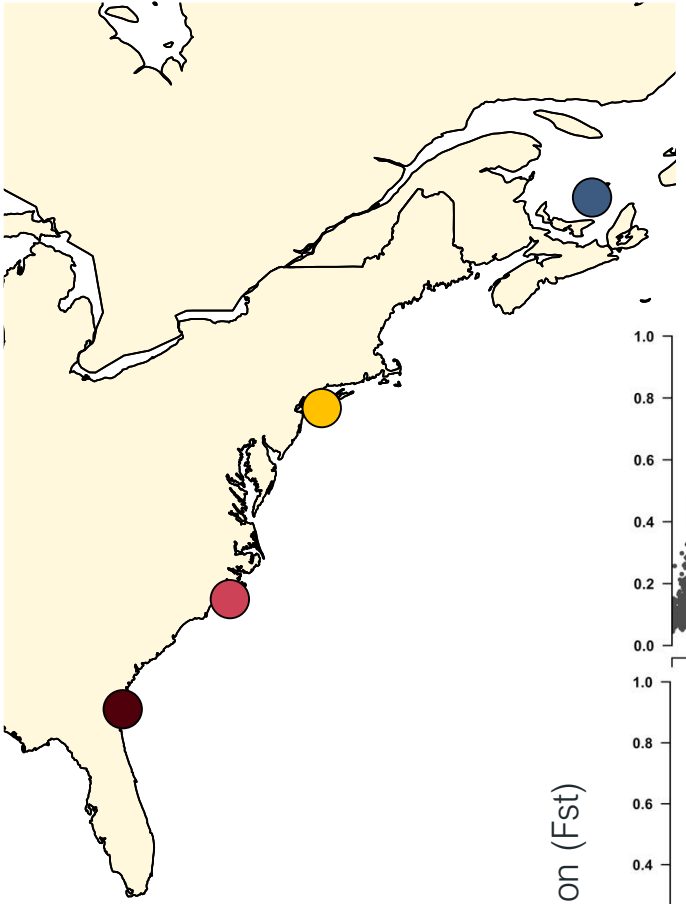


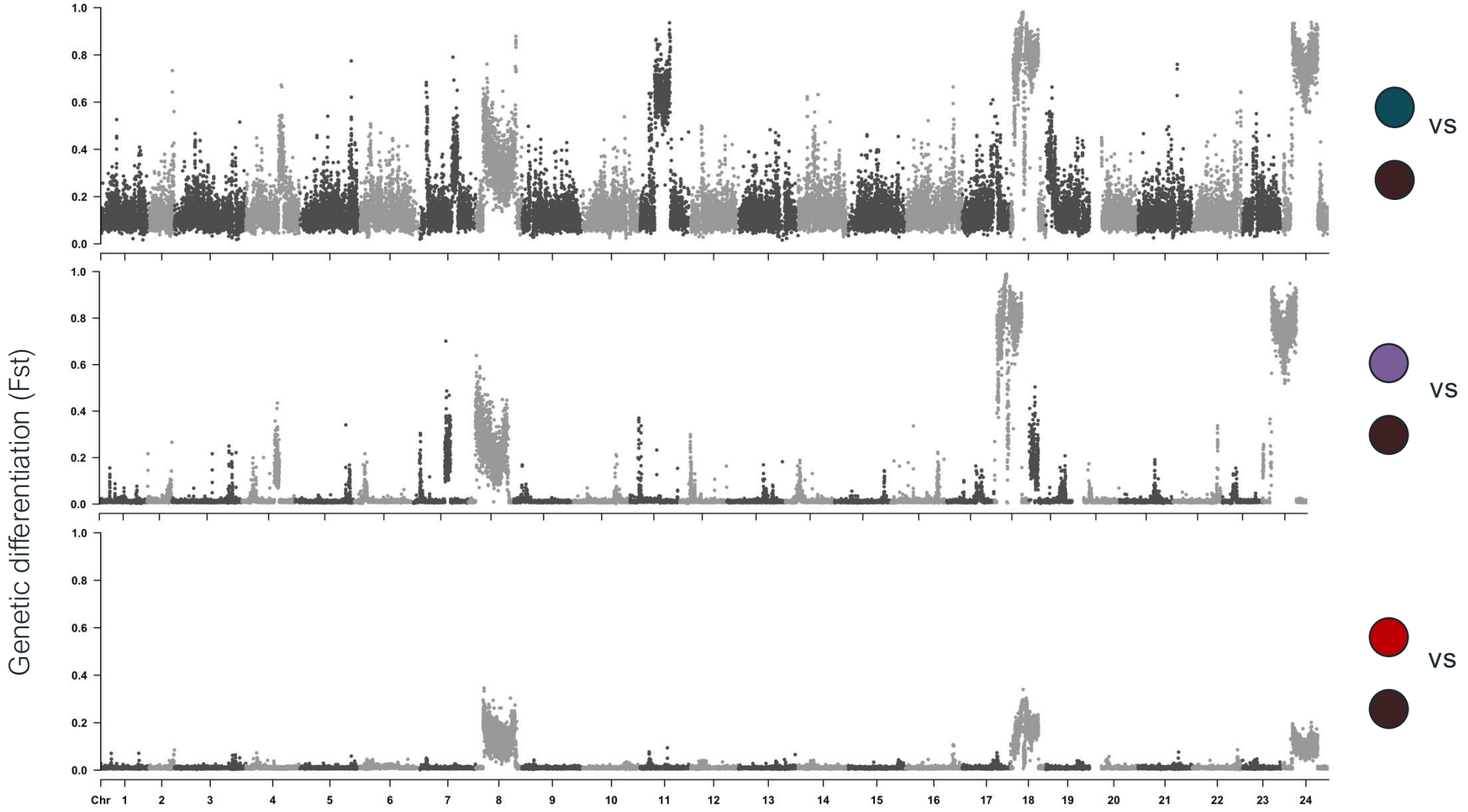
LD decay

- Can be used to reconstruct population histories
- Determine size of sliding windows
- Prune SNPs for population structure analyses

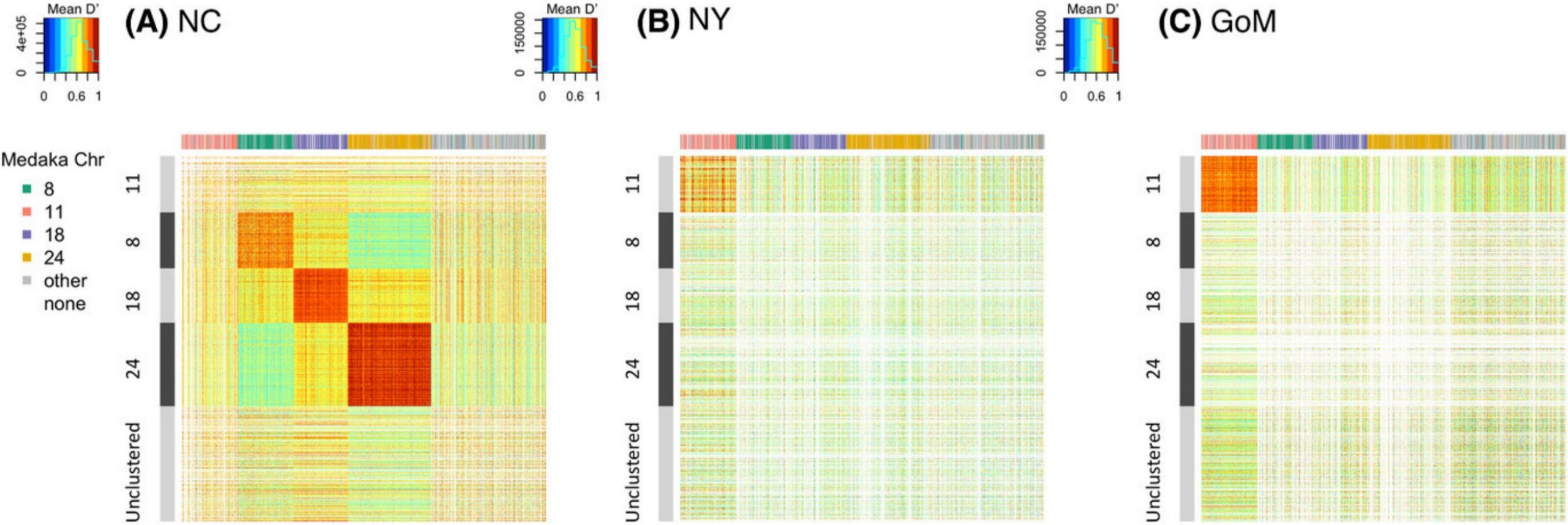


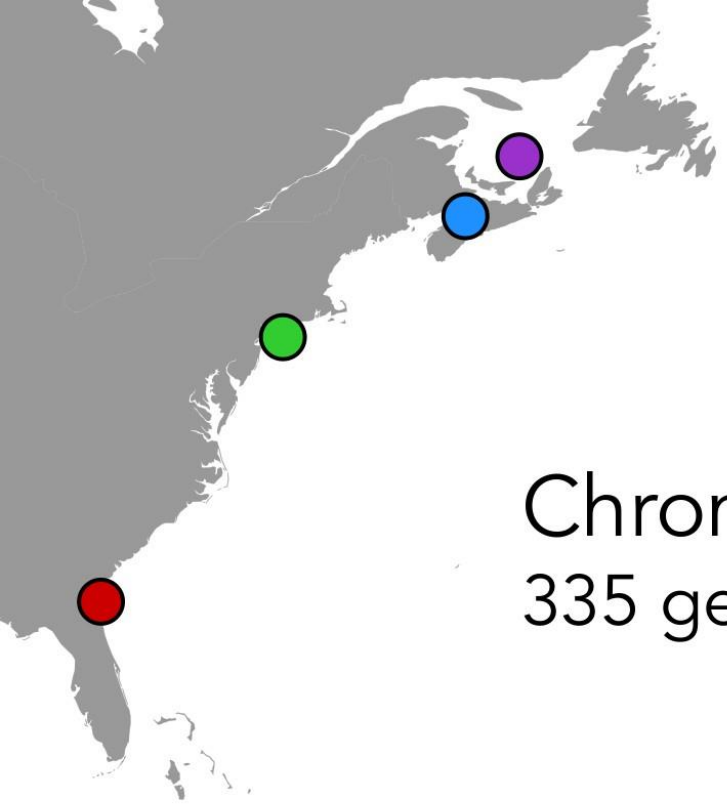
Heterogeneous gene flow and divergence patterns across latitudes

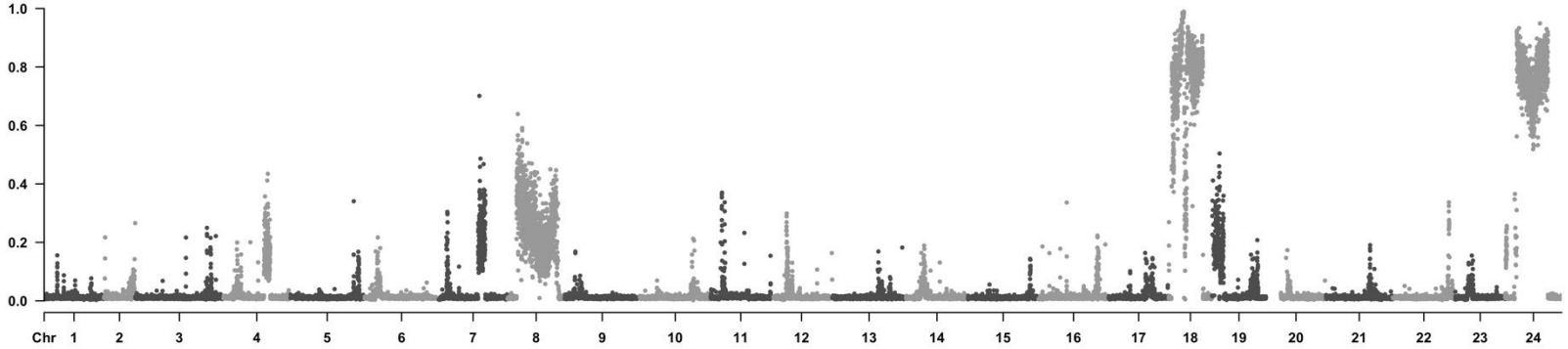




LD patterns among outliers

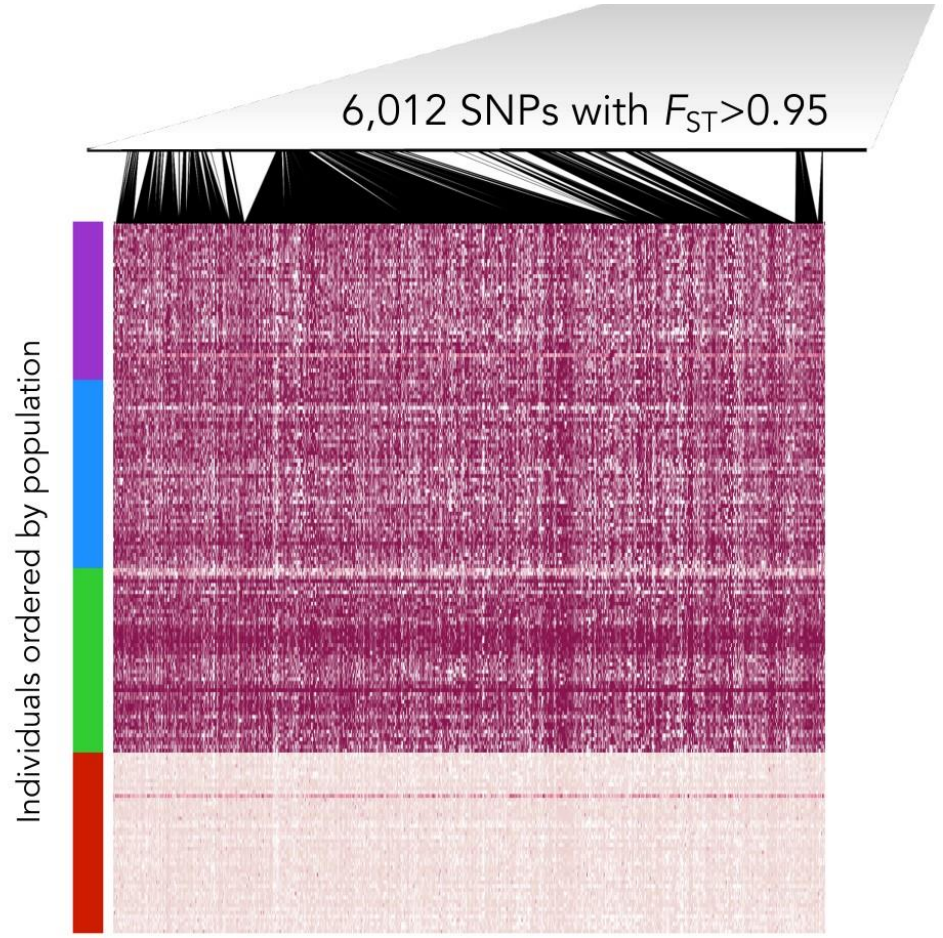






Chromosome 24

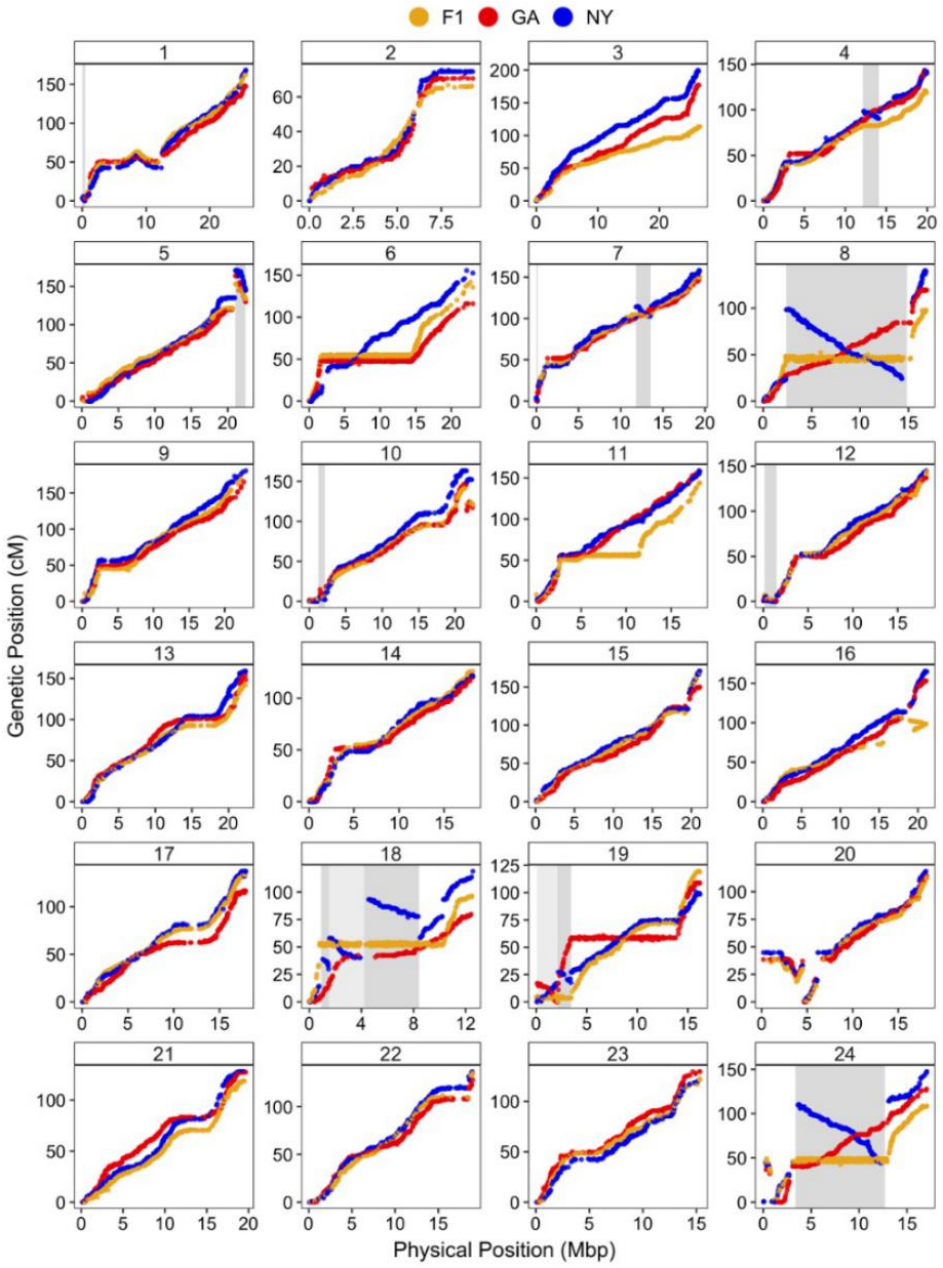
335 genes



Most likely genotype for each individual

- Northern homozygote
- Heterozygote
- Southern homozygote

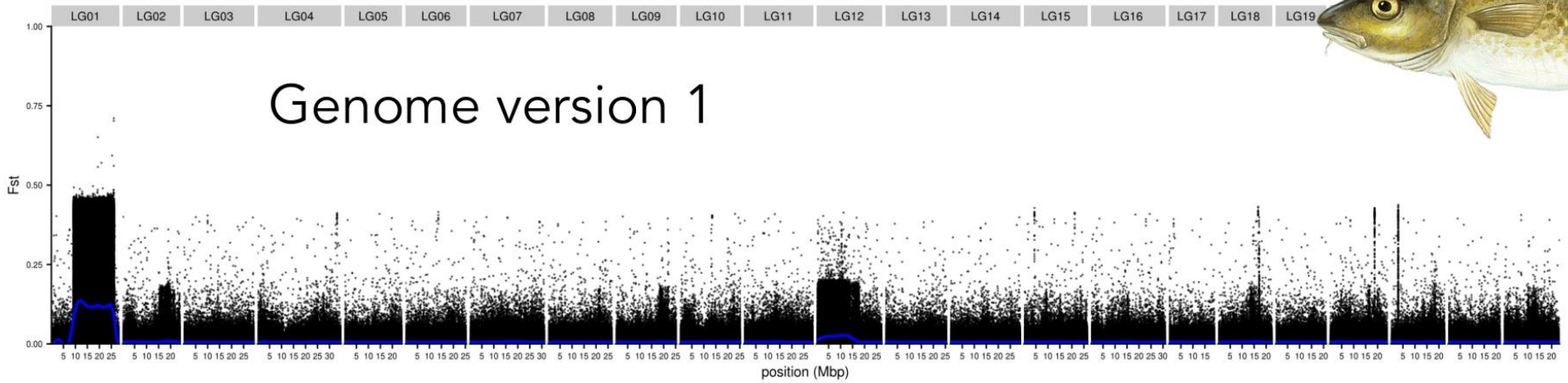
Linkage mapping confirms presence of massive inversions

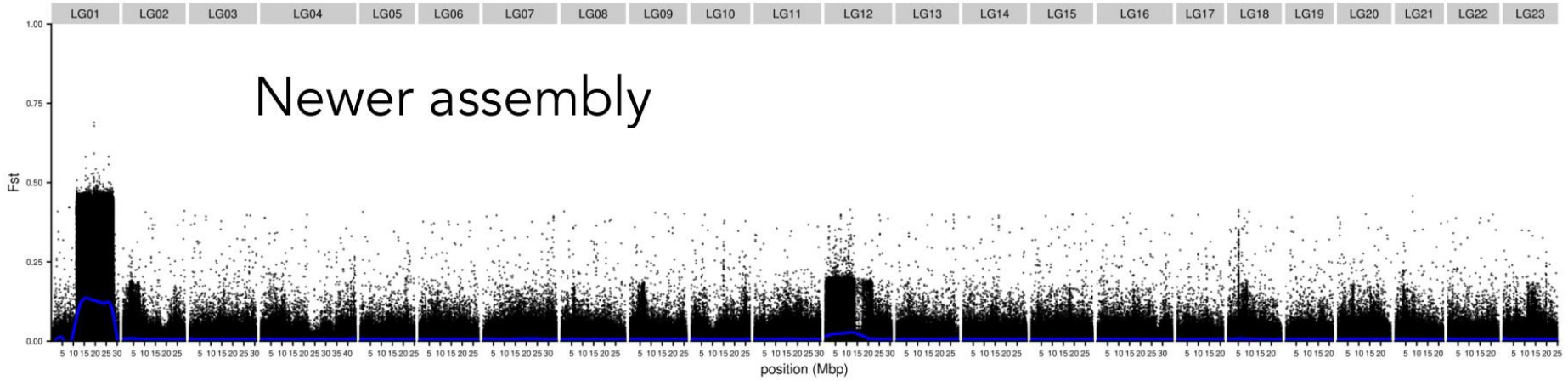


Akopyan et al. 2022

Examples from Atlantic cod



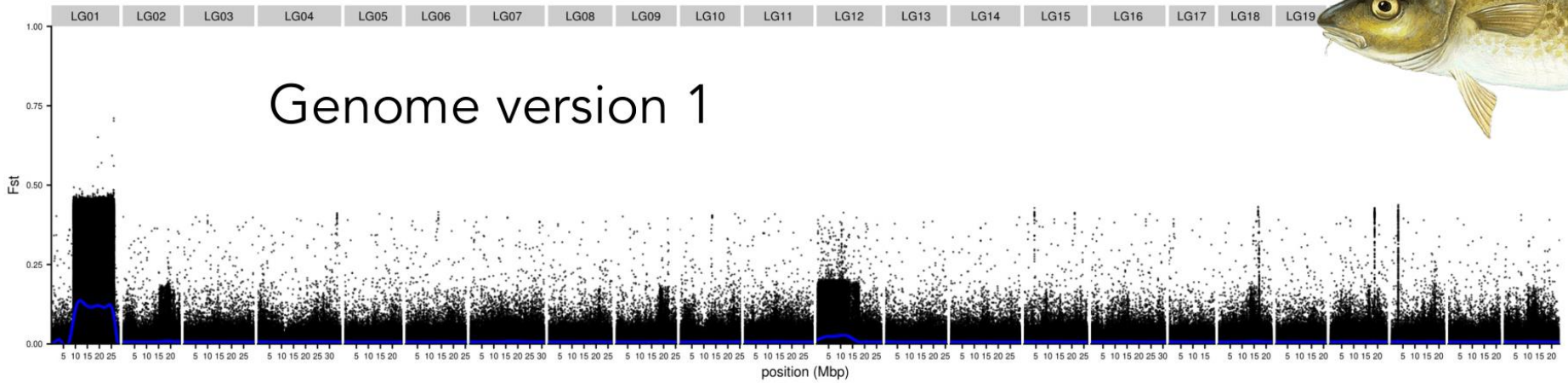


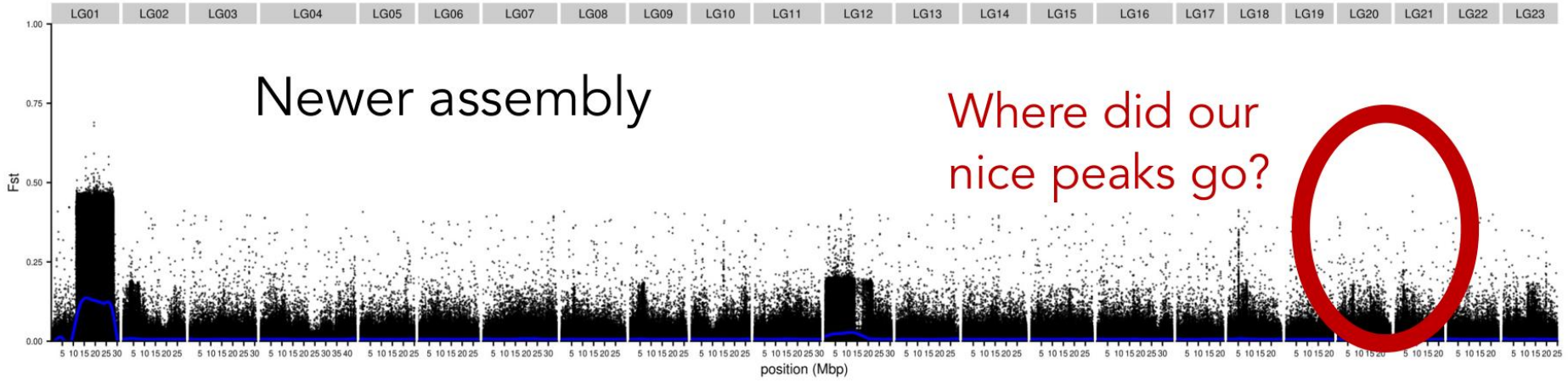


Nicolas Lou, unpubl.

Examples from Atlantic cod

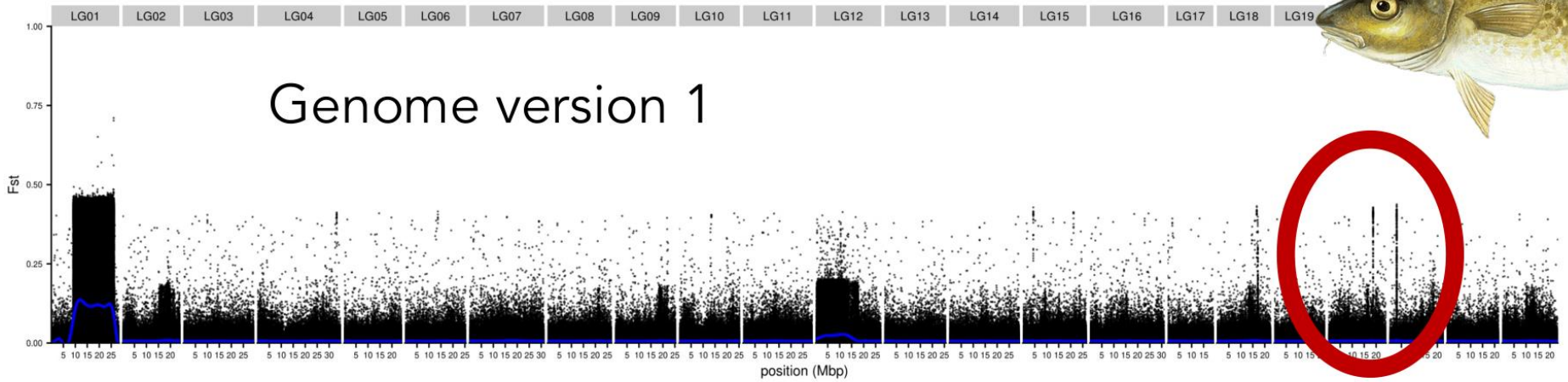


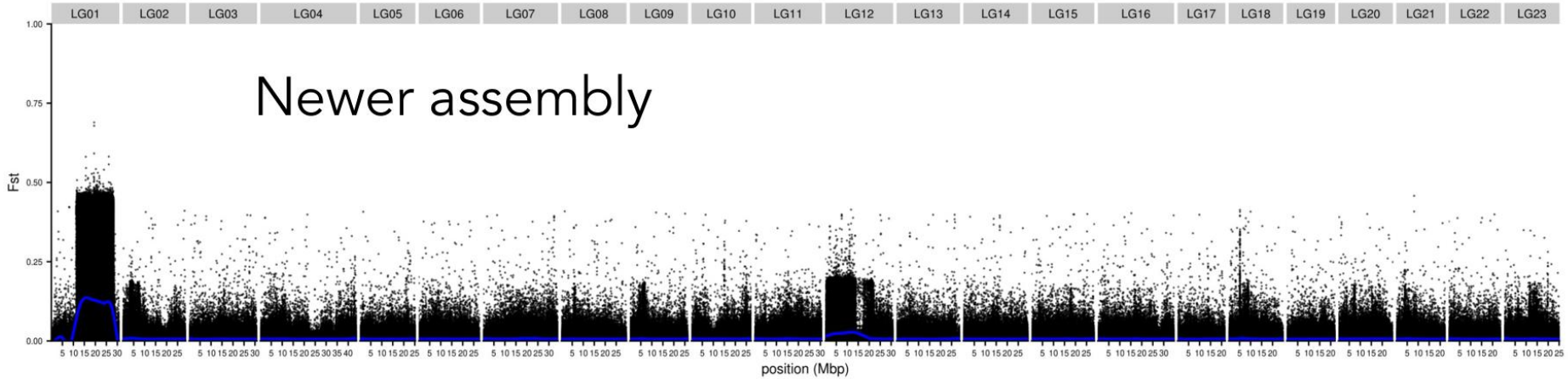




Nicolas Lou, unpubl.

Examples from Atlantic cod





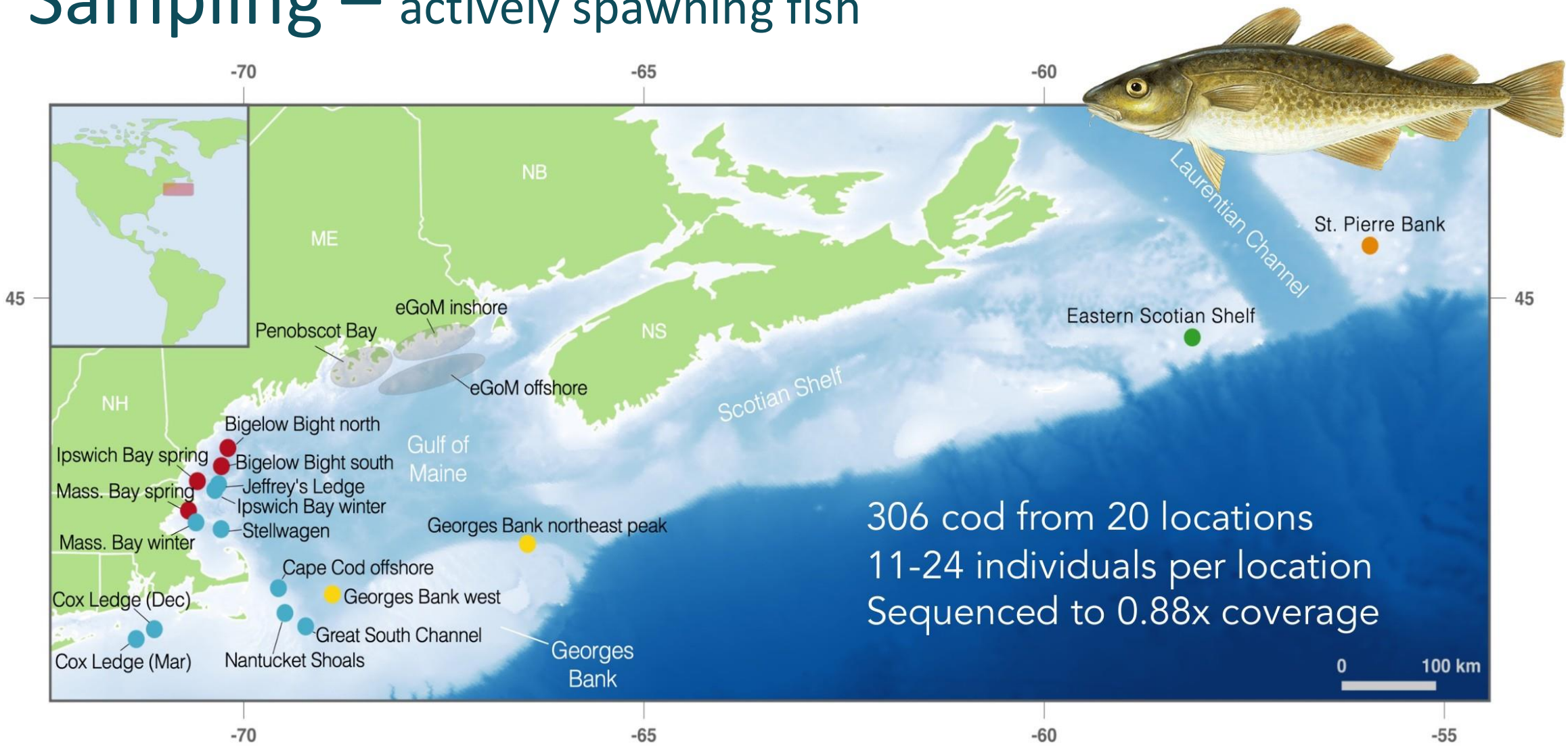


Are the isolated peaks in LD with SNPs in the inversions?

Nicolas Lou, unpubl.

LD can also affect inference of population structure

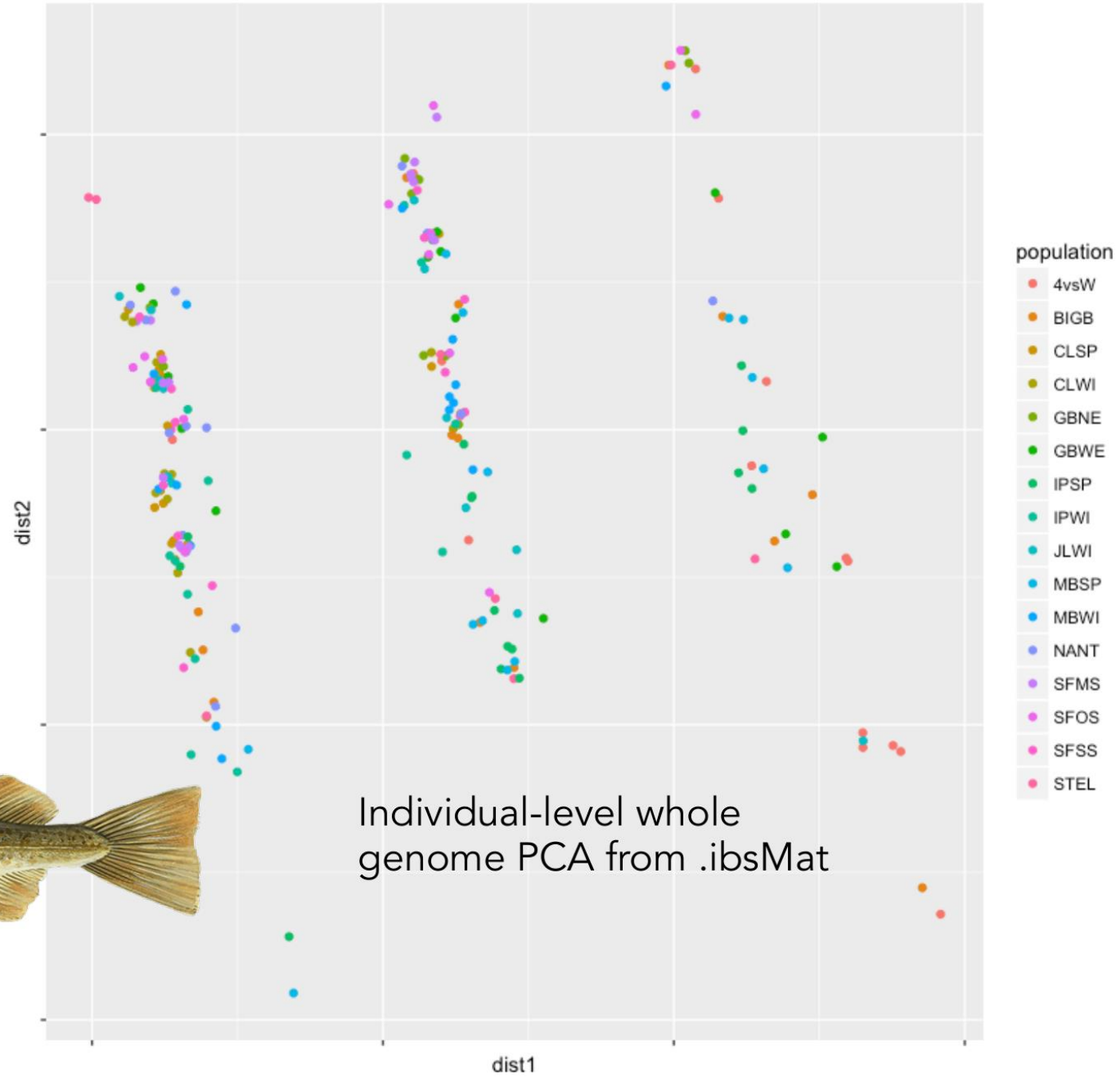
Sampling – actively spawning fish



PCA

- What could explain this pattern?

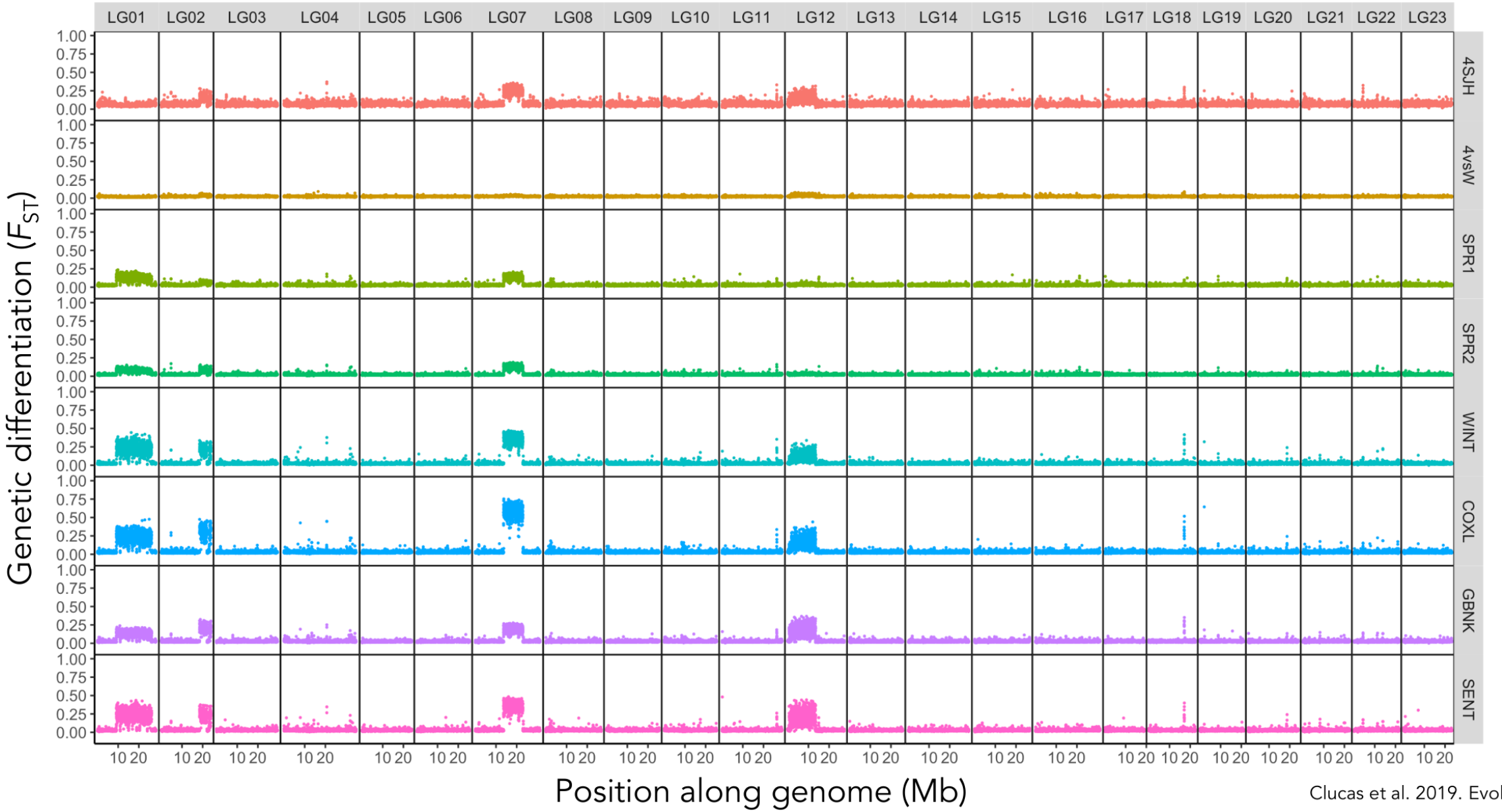




Individual-level whole genome PCA from .ibsMat

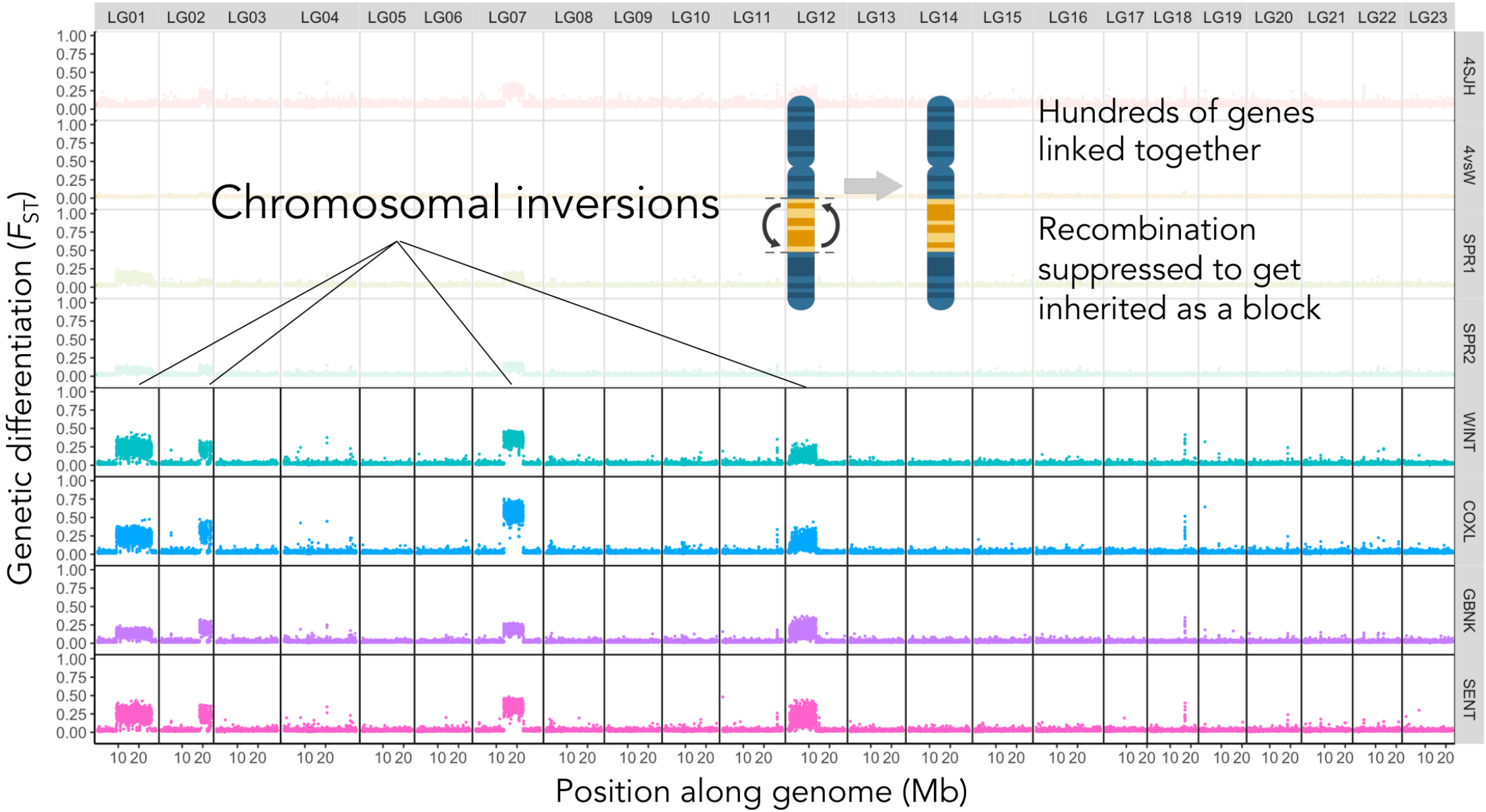
Gemma Clucas and Nicolas Lou (unpublished results)

Pairwise comparisons to St. Pierre Bank, Canada (15 kb windows)

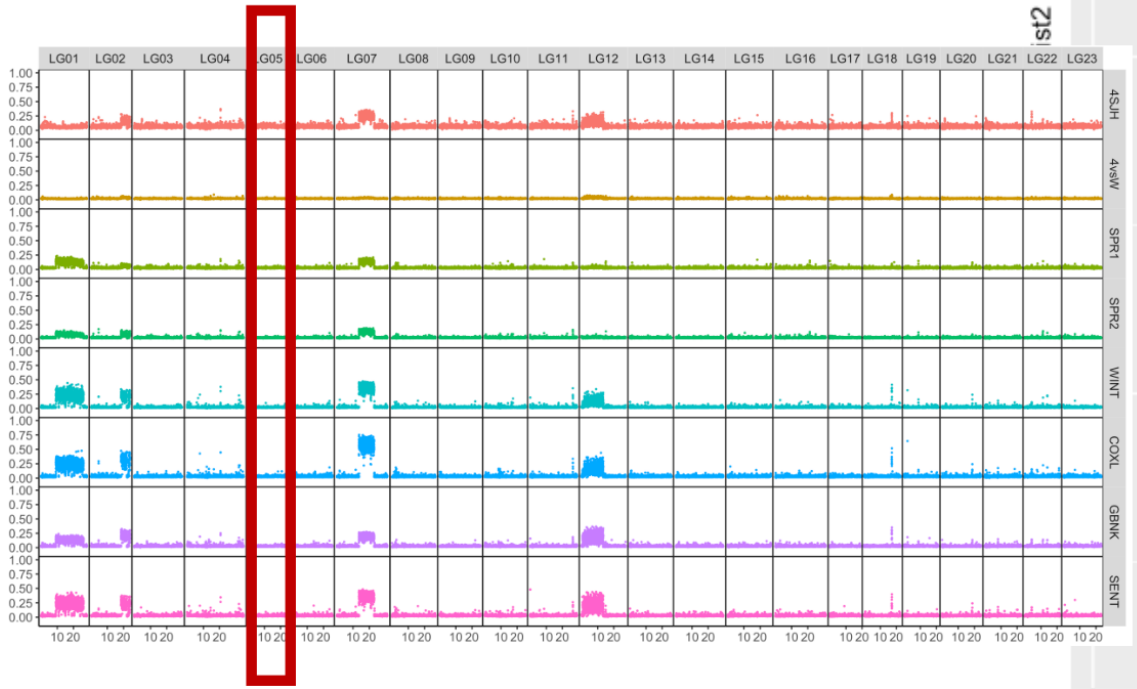


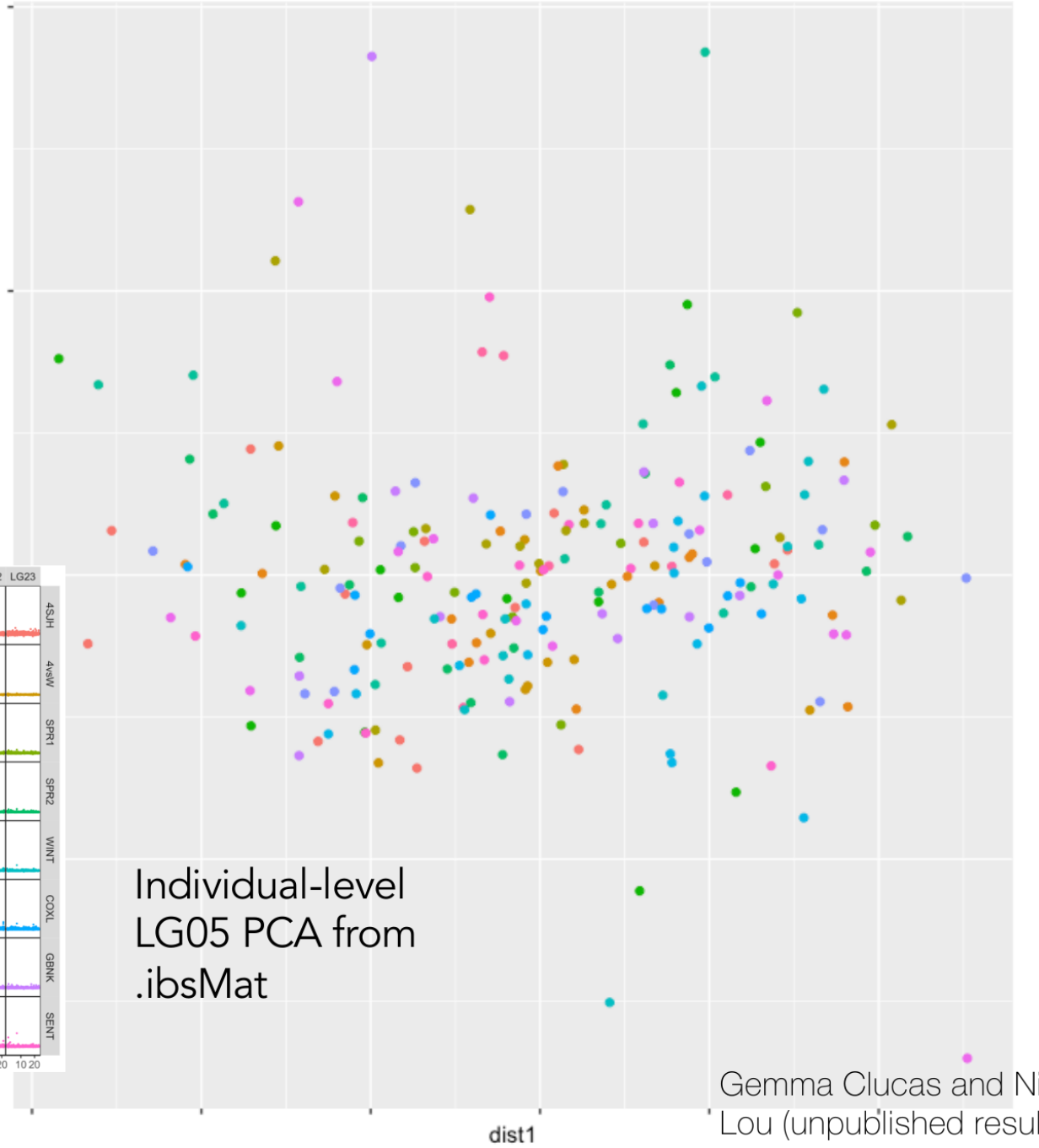
Clucas et al. 2019. Evol. Appl.

Pairwise comparisons to St. Pierre Bank, Canada (15 kb windows)



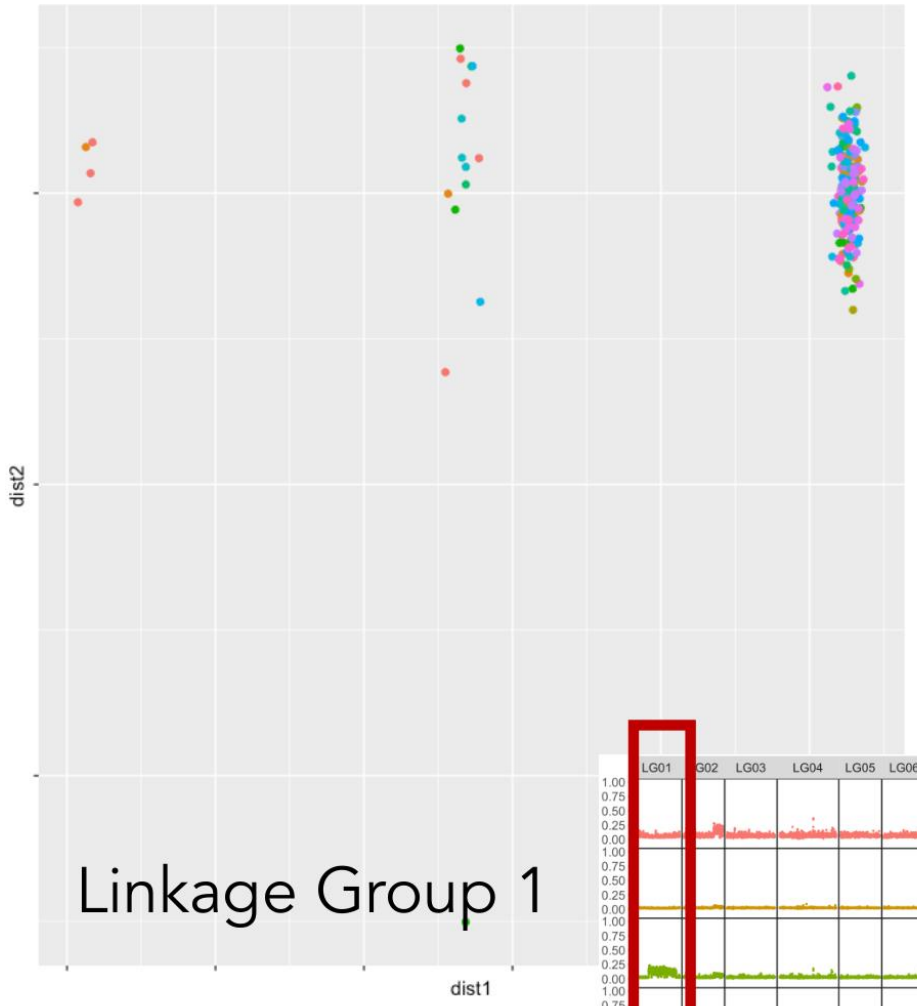
PCA with SNPs only on Linkage Group 5

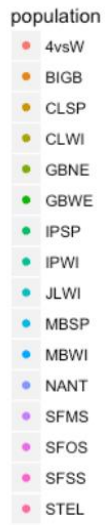




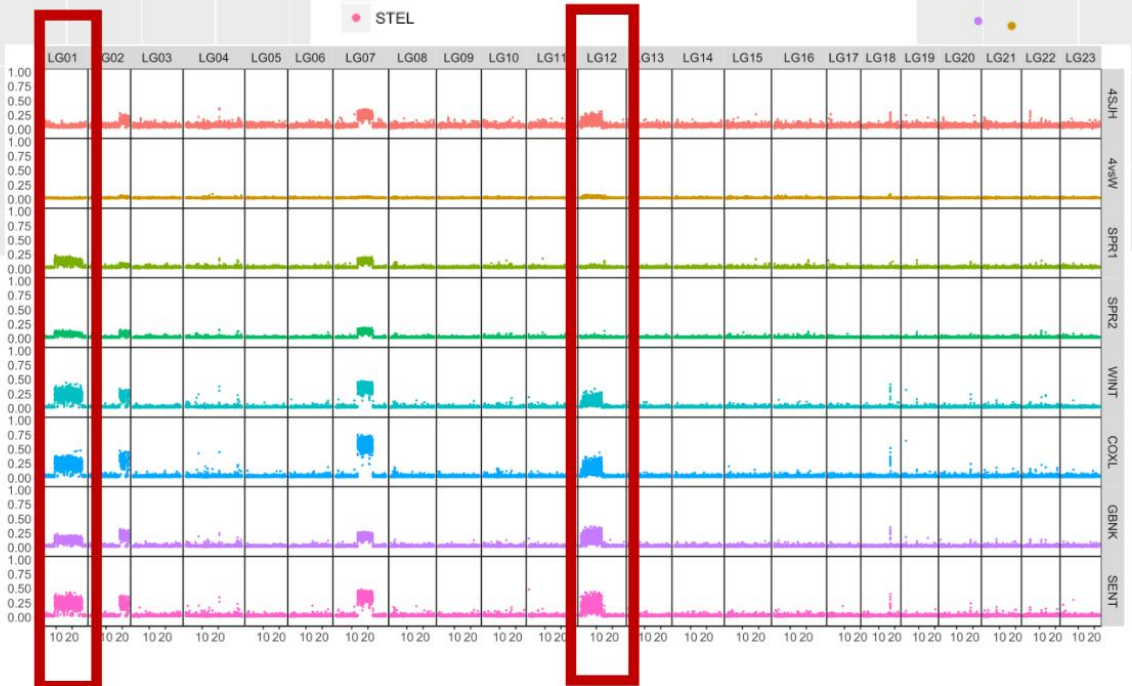
Individual-level LG05 PCA from .ibsMat

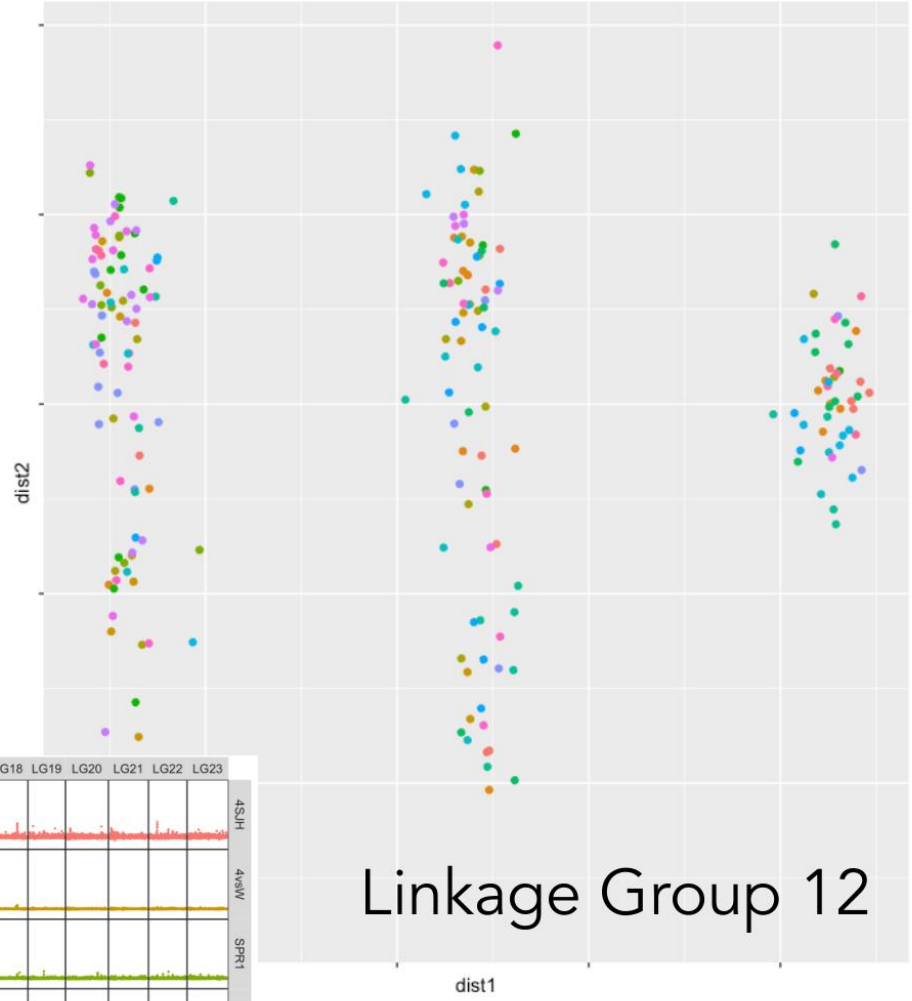
Gemma Clucas and Nicolas Lou (unpublished results)

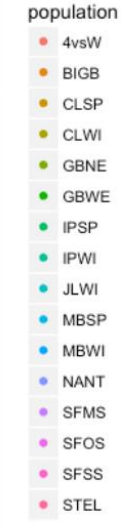




Linkage Group 1



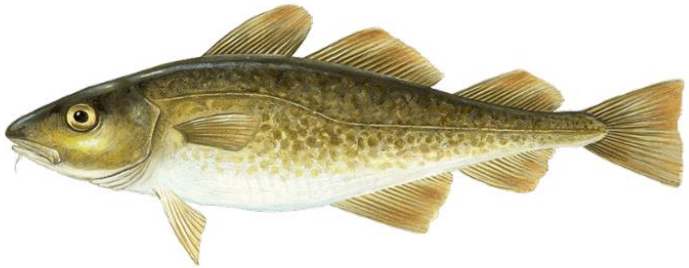




Linkage Group 12

Plots of .ibsMat

Gemma Clucas and Nicolas Lou
(unpublished results)

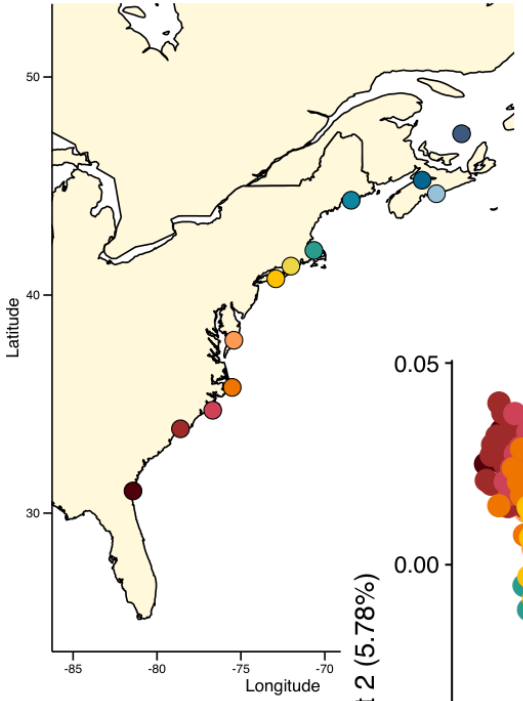


The need for LD pruning data sets

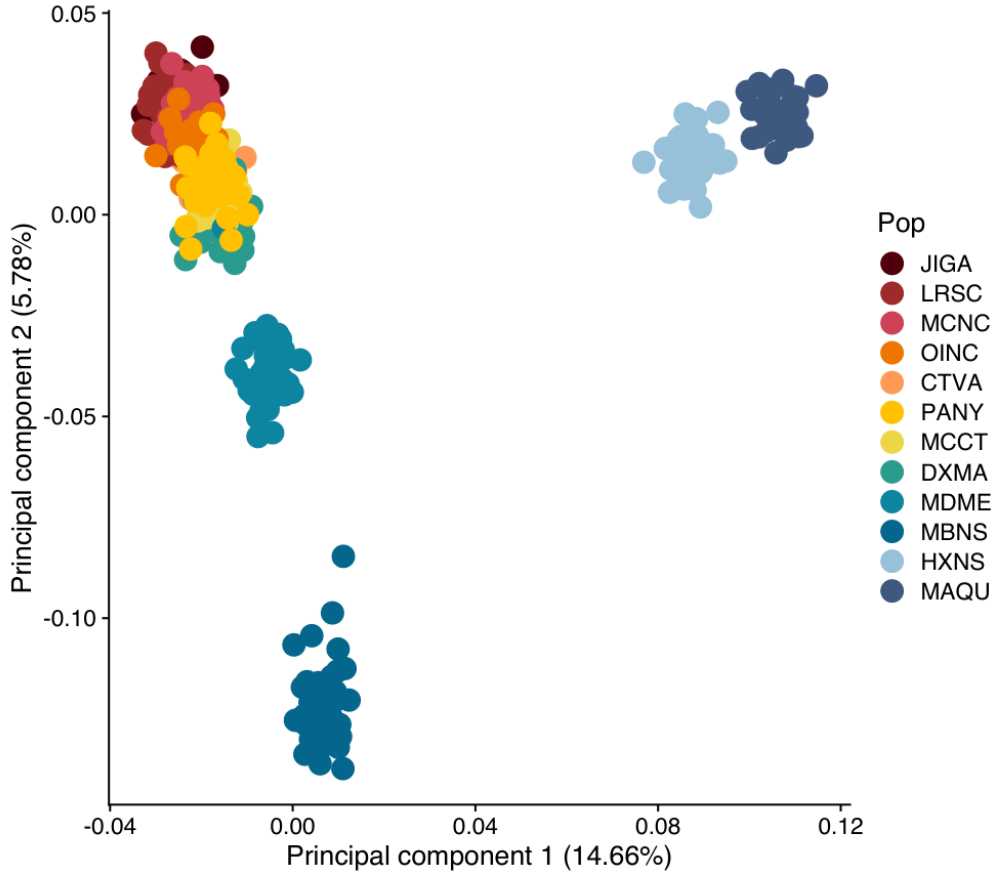
- Strongly linked regions are inherited as a single block, i.e. are effectively a single genetic marker
- If it contains many SNPs, this “single marker” will be way over-represented in analyses of geographic variation
- Many downstream analysis approaches assumes independence among markers

Another example – inversions don't always result in clustering into three genotype classes

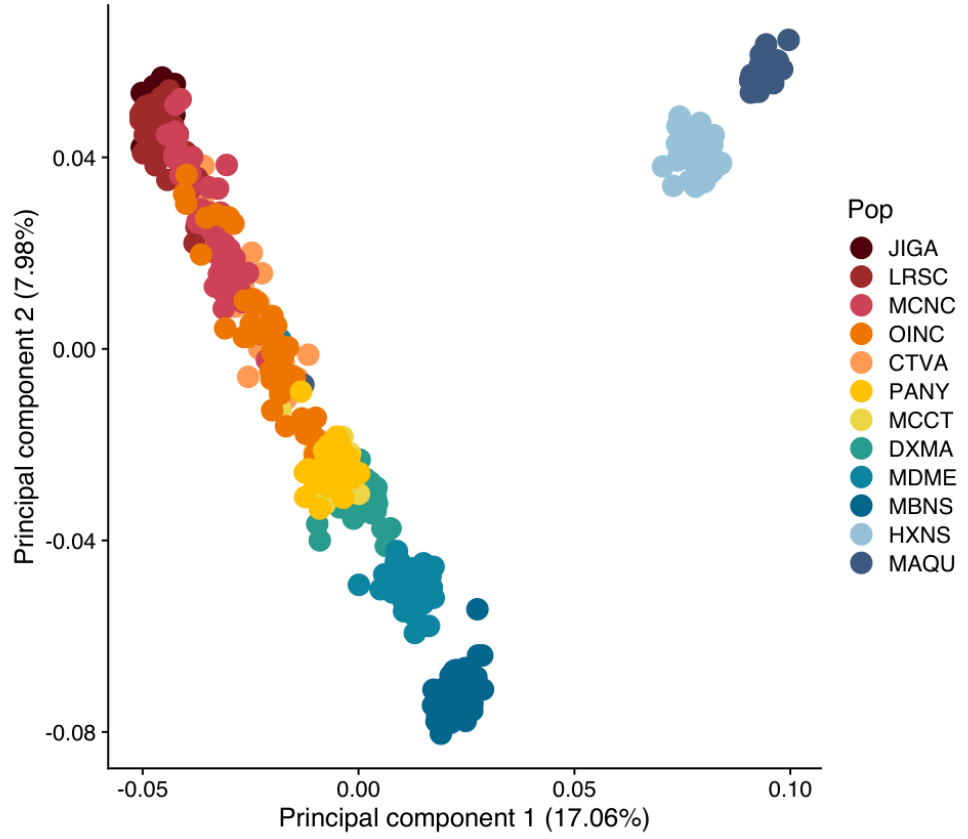
Arne Jacobs (unpublished results)



Without inversions



With inversions



Bioinformatics, 35(19), 2019, 3855–3856
doi: 10.1093/bioinformatics/btz200
Advance Access Publication Date: 23 March 2019
Applications Note

OXFORD

Genetics and population analysis

ngsLD: evaluating linkage disequilibrium using genotype likelihoods

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[1]Department of Life Sciences, Silwood Park Campus, Imperial College London, Ascot, UK, [2]Department of Animal and Plant Sciences, University of Sheffield, Sheffield, UK and [3]Center for GeoGenetics, Natural History Museum of Denmark, University of Copenhagen, Copenhagen, Denmark

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Associate Editor: Oliver Stegle

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Abstract

Motivation: Linkage disequilibrium (LD) measures the correlation between genetic loci and is highly informative for association mapping and population genetics. As many studies rely on called genotypes for estimating LD, their results can be affected by data uncertainty, especially when employing a low read depth sequencing strategy. Furthermore, there is a manifest lack of tools for the analysis of large-scale, low-depth and short-read sequencing data from non-model organisms with limited sample sizes.

Results: *ngsLD* addresses these issues by estimating LD directly from genotype likelihoods in a fast, reliable and user-friendly implementation. This method makes use of the full information available from sequencing data and provides accurate estimates of linkage disequilibrium patterns

Downloaded from https://academic.oup.com/bioinformatics/article/35/19/3855/5418793

ngsLD

ngsLD estimates the four two-locus haplotype frequencies (p_AB, p_Ab, p_aB, p_ab) by EM.

The EM integrates over both genotype uncertainty (the genotype likelihoods) and phase uncertainty, since double heterozygotes cannot be resolved. This assumes the two haplotypes in each individual combine at random, i.e. HWE in the sample.

It estimates:

1. r^2_ExpG: correlation of expected genotypes
2. D
3. D’
4. r^2

ngsLD: comparison of statistics

	r^2_ExpG	D	D’	r^2 (EM)
Level	Genotype	Haplotype	Haplotype	Haplotype
Estimation	Pearson on posterior mean dosages	EM	EM	EM
Range	0 to 1	Depends on allele frequencies; arbitrary sign	−1 to 1	0 to 1, capped when MAFs differ
Key assumption	None on gamete union	HWE	HWE	HWE
Main use	Quick screen; robust when HWE is violated	Building block	Recombination history, haplotype blocks	Pruning, LD decay, tagging, N_e
Main weakness	Shrinkage at low depth	Not comparable between pairs	Inflated by rare alleles and small n	Depends on MAF; affected by structure through the HWE assumption